

Winding Ambiguity & Specular Glint Analysis

Weekly Report — 13 March 2026

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Context

Test case: IS-901. $\omega_{\text{true}} = (0.5, -0.3, 2.0)^\circ/\text{s}$, $|\omega_{\text{true}}| = 2.083^\circ/\text{s}$.

Experiments 1–4: 120 s window, 100 observations, $\Delta t_{\text{samp}} \approx 1.21$ s, $\sigma = 0.05$ mag.

Experiments 5–6: 3600 s window, 500 observations, peaks at indices 183, 260, 360.

Experiments 7–10: 3600 s window, 500 observations. Exp 8 uses 30 random (q_0, ω_0) pairs.

Experiments (Recap from 10 March)

Experiment 1: Forward Propagation — Analytic ω , Exact Attitude, Hi-fi

Variables	None (no free parameters)
Constraints	IS-901; q_{k-1}, q_k exact from true trajectory; orientation: arb160; hi-fi shadowing
Residual	MSE between reconstructed and observed LC (computed, not optimised)
Solver	None — $\omega = \text{rotvec}(q_{k-1}^{-1} q_k) / \Delta t_{\text{samp}}$ (analytic); Euler forward propagation from q_k

Result: Reconstructed LC overlays truth. At $\Delta t_{\text{samp}} \approx 1.21$ s the rotation is $\sim 2.5^\circ$ —no winding ambiguity.

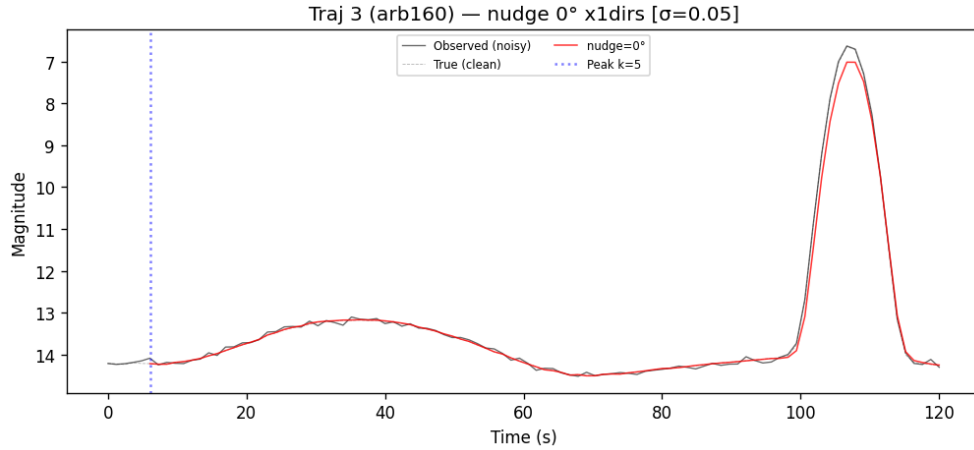


Figure 1: Hi-fi forward propagation from exact q_k with analytic ω (trajectory 3, arb160).

Experiment 2: Forward Propagation — Analytic ω , 3° Attitude Nudge, Hi-fi

Variables	Nudge direction (5 random unit vectors, seed 123; not optimised)
Constraints	Same analytic ω as Exp 1; q_k nudged 3° before propagation; hi-fi
Residual	MSE per direction (computed, not optimised)
Solver	None — same analytic ω ; Euler forward propagation from nudged q_k

Result: Anisotropic sensitivity. Near the glint, some directions shift the peak by >2 mag; others are benign.

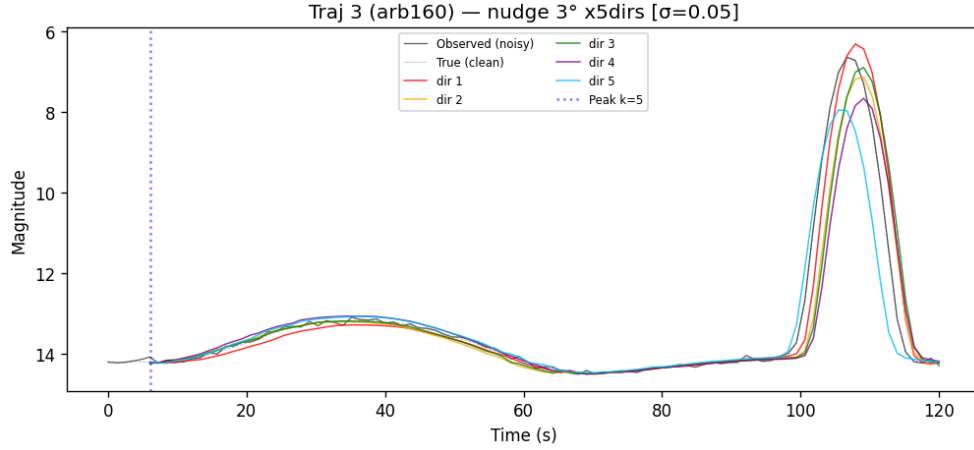


Figure 2: Hi-fi forward propagation from 3°-nudged q_k in 5 directions (trajectory 3, arb160).

Experiment 3: Forward Propagation — Lo-fi Reconstruction, Exact Attitude

Variables	None
Constraints	Same analytic ω and exact q_k as Exp 1; lo-fi reconstruction (no self-shadowing); observed LC is hi-fi; orientation: arb160
Residual	MSE(lo-fi reconstruction, hi-fi observed)
Solver	None — forward evaluation only

Result: Lo-fi phantom glint at $t \approx 60$ s—facet in self-shadow produces false peak.

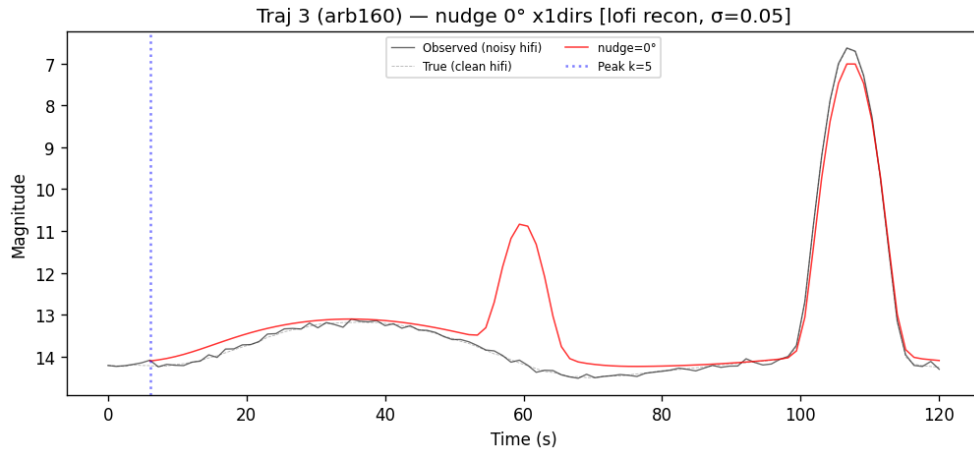


Figure 3: Lo-fi reconstruction vs hi-fi observed (trajectory 3, arb160). Phantom peak at $t \approx 60$ s.

Experiment 4: Forward Propagation — Lo-fi, Different Orientation

Variables	Nudge direction (5 random unit vectors)
Constraints	Same analytic ω ; q_k nudged 3° ; lo-fi reconstruction; orientation: 30° about $\hat{x}$
Residual	MSE(lo-fi, hi-fi) per direction
Solver	None — forward evaluation only

Result: No phantom peak at this orientation. Lo-fi failure depends on active self-shadowing, unpredictable a priori.

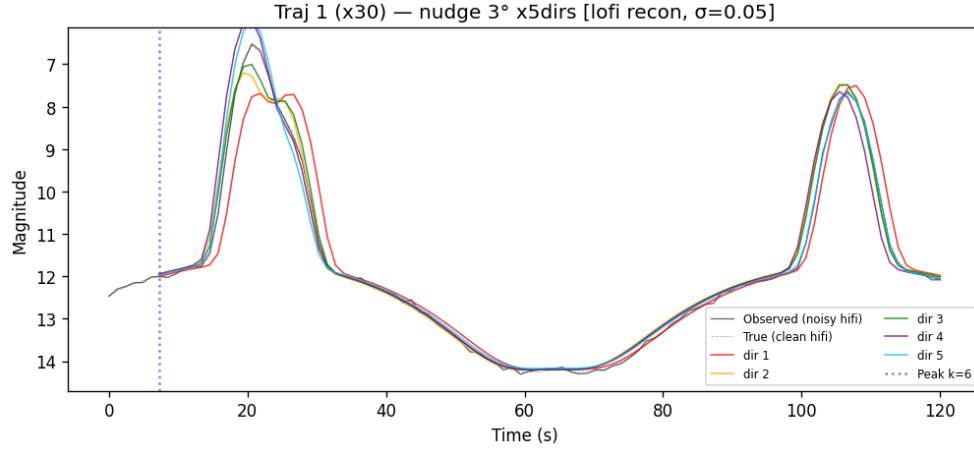


Figure 4: Lo-fi at benign orientation (trajectory 1, x30). No phantom peak.

Experiment 5: Staircase ω Enumeration

Variables	$\omega \in \mathbb{R}^3$ (optimised per step)
Constraints	q_A, q_B exact (oracle) at peaks 183, 260; $\Delta t = 555.5$ s; soft lower barrier on $ \omega ^2$ per step
Residual	Objective: $1 - (q_{\text{prop}} \cdot q_B)^2 + \lambda_{\text{barrier}}$; post-hoc trough: $ b_{\text{pred}} - b_{\text{ref}} $ (lo-fi, single epoch)
Solver	L-BFGS-B, 8 sequential steps; axis-angle init step 0; $+2\pi/\Delta t$ increment subsequent steps

Result: 8 winding solutions found. Trough scoring selects step 2 (wrong)—insufficient temporal leverage.

Step	$ \omega $ ($^\circ/\text{s}$)	Arrival err	Trough err
0	0.316	0.0	1.719
1	0.959	0.0	0.099
2	1.601	0.0	0.021 $\leftarrow$ best trough
3	2.230	0.0	0.792 $\leftarrow$ true
4	2.883	0.036	0.044
5	3.509	0.0	0.134
6	4.180	0.0	1.749
7	4.824	0.0	1.491

Experiment 6: Angular Momentum Conservation Filter — Two-Leg Staircase

Variables	$\omega \in \mathbb{R}^3$ per leg per step
Constraints	q_A, q_{mid}, q_B exact at peaks 183, 260, 360; leg 0 $\Delta t = 555.5$ s; leg 1 $\Delta t = 721.4$ s; lower + upper $ \omega $ barriers ($\pm 0.8 \delta$)
Residual	Per step: $1 - (q_{\text{prop}} \cdot q_{\text{end}})^2 + \text{barriers}$; L-filter: $\ \mathbf{L}_{\text{leg } 0}[k] - \mathbf{L}_{\text{leg } 1}[j]\ $
Solver	L-BFGS-B per step; exhaustive $8 \times 8 = 64$ L-comparison at shared peak 260

Result: Selected ($k=0, j=0$). True: ($k=3, j=1$). **Incorrect**—leg 1 staircase broken (near- π singularity skips bands).

Step	Leg 0 $ \omega $ ($^\circ/\text{s}$)	Leg 1 $ \omega $ ($^\circ/\text{s}$)
0	0.316	0.249
1	0.959	3.279 $\leftarrow$ gap
2	1.601	3.509
3	2.230	3.995

Decisions from 10 March

- MIP ruled out (consulted Jack—insufficient discrete structure).
- Lo-fi scoring unreliable near glints (Exp 3–4).
- Single-trough scoring insufficient (Exp 5).
- L-conservation most promising ω filter, pending staircase fix.

New Experiments (This Week)

Experiment 7: PAB Alignment Diagnostic at Brightness Peaks

Variables	None (diagnostic—no free parameters)
Constraints	IS-901, standard test case; 500 epochs / 3600 s; hi-fi with per-facet flux decomposition (<code>animate=True</code>); 3840 facets grouped into 14 unique normal directions (rounded to 4 dp); peaks detected via <code>argrelmin(mag, order=5)</code>
Residual	$\hat{\mathbf{n}}_g \cdot \widehat{\text{PAB}}_{\text{body}}$ for each group g at each epoch; fractional flux = $\Sigma f_g / \Sigma f_{\text{total}}$ per group at peaks
Solver	None — forward evaluation + decomposition

Result: 43 peaks detected. At all 11 bright peaks ($\text{mag} < 9$): a single normal group captures $>77\%$ of total flux with $\hat{\mathbf{n}} \cdot \widehat{\text{PAB}} > 0.99$. Bright peaks are specular glints. Dim peaks ($\text{mag} > 11$) are diffuse—dominated by bus broadside area, no near-perfect alignment.

Micro-34: PAB Alignment Diagnostic at Lightcurve Peaks

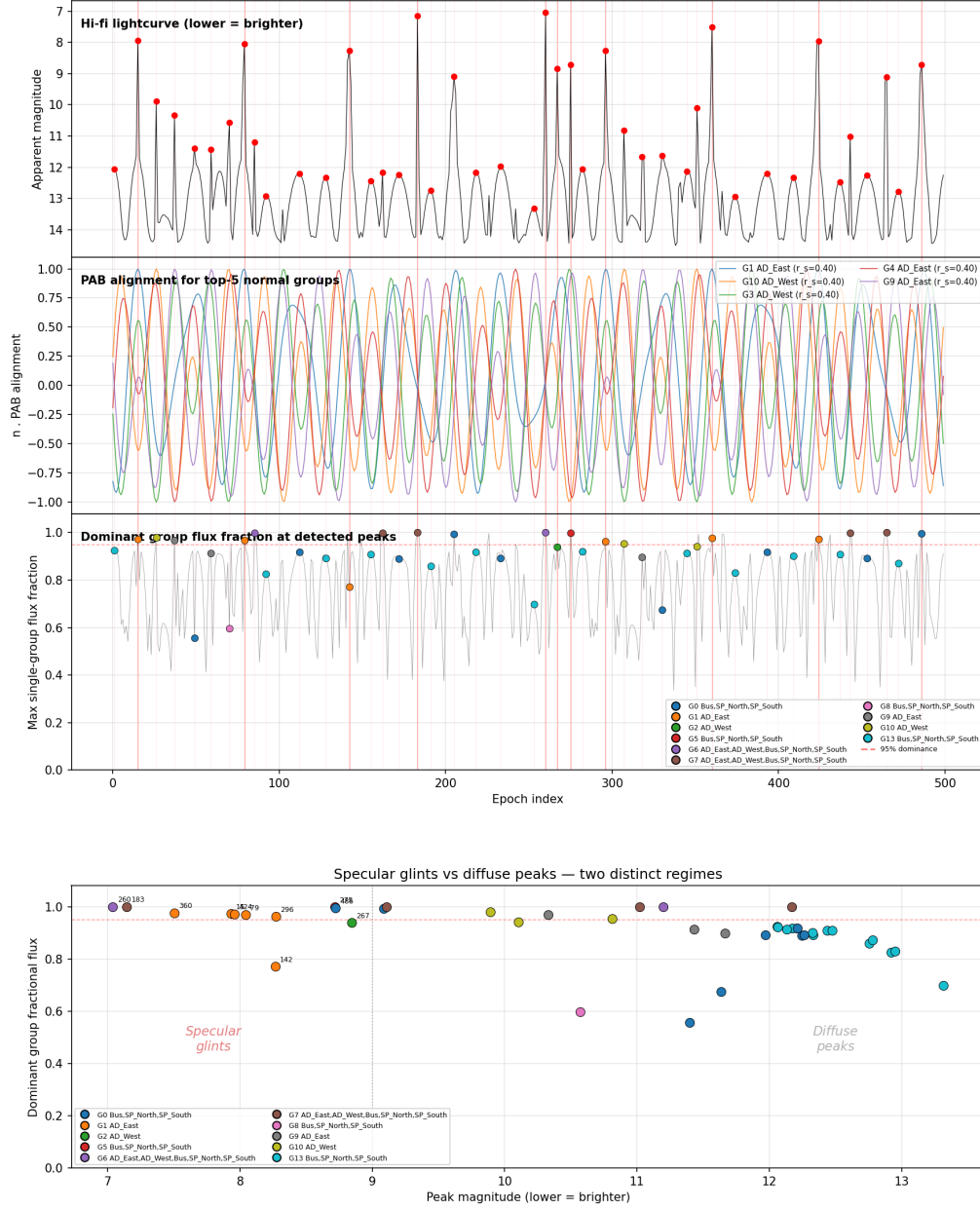


Figure 5: micro34: 4-panel PAB diagnostic. Top: lightcurve. Second: $\hat{n} \cdot \text{PAB}$ traces. Third: dominant-group flux fraction at peaks. Bottom: two-regime scatter (specular vs diffuse).

Experiment 8: Multi-Trajectory Glint Robustness (30 random trajectories)

Variables	None (statistical survey)
Constraints	30 random (q_0, ω_0) pairs; q_0 uniform on $\text{SO}(3)$; $ \omega $ uniform in $[0.5, 5.0]^\circ/\text{s}$; seeds 0–29; same IS-901 geometry/SPICE; hi-fi with per-facet flux; peaks via <code>argrelmin(order=5)</code>
Residual	Per bright peak ($\text{mag} < 9$): classified “specular glint” if dominant group frac > 0.77 and $\hat{\mathbf{n}} \cdot \text{PAB} > 0.99$; else “near-miss”
Solver	None — forward evaluation per trajectory (~ 58 s each, 29 min total)

Result:

Total bright peaks (all 30 trajectories)	439
Specular glints (strict)	392 (89.3%)
Near-misses	47 (10.7%)
Genuine counterexamples	0

All 47 near-misses have $\hat{\mathbf{n}} \cdot \text{PAB} > 0.984$ and $f > 0.54$ —boundary cases, not violations. Near-miss rate increases with $|\omega|$ (2% at $< 1^\circ/\text{s}$; 14% at $3\text{--}5^\circ/\text{s}$). Glint count $\approx 2.9|\omega| + 6.3$ (linear).

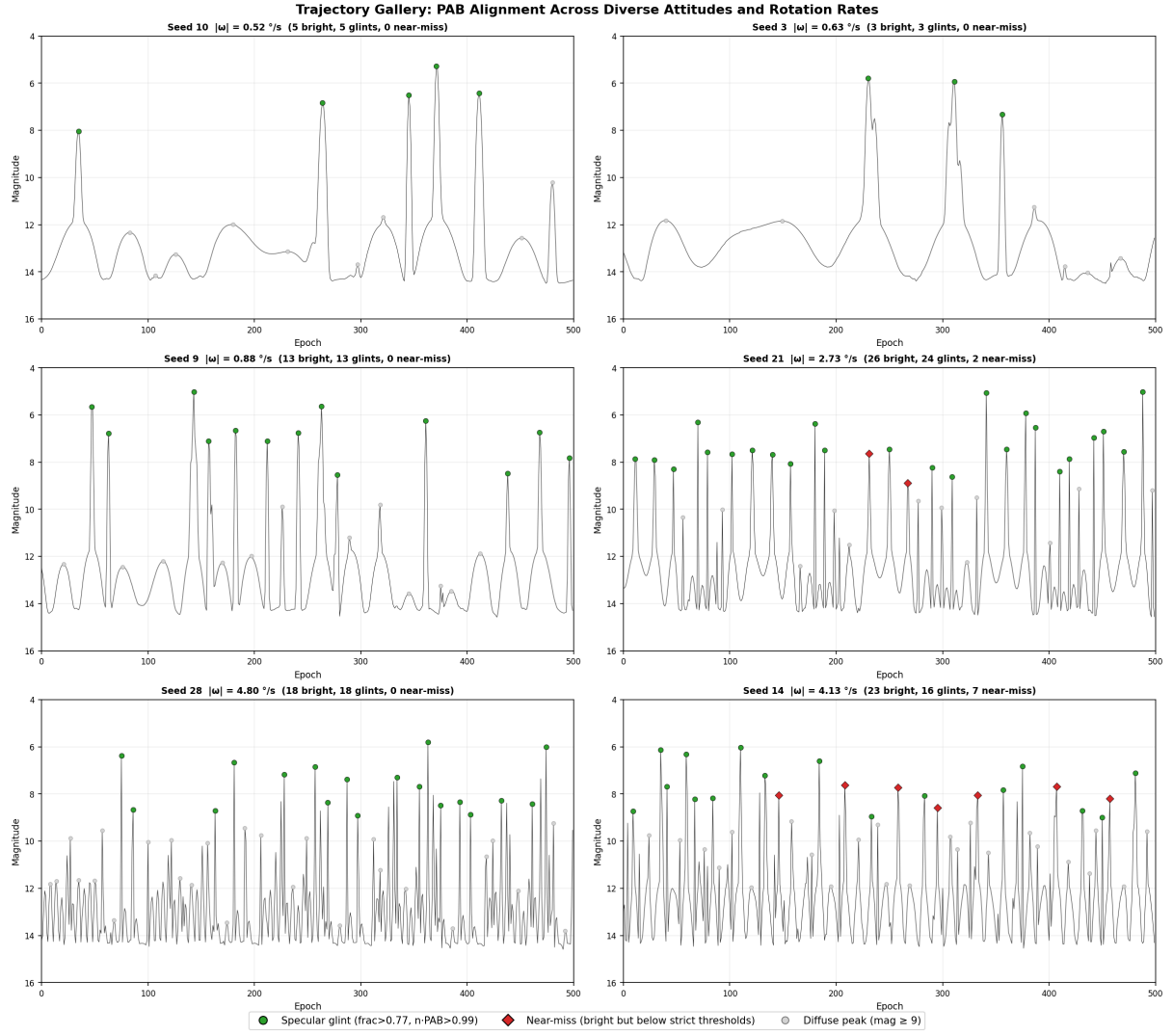


Figure 6: micro35: 6 representative trajectories ($|\omega| = 0.52\text{--}4.80^\circ/\text{s}$). Green = specular glint, red = near-miss.

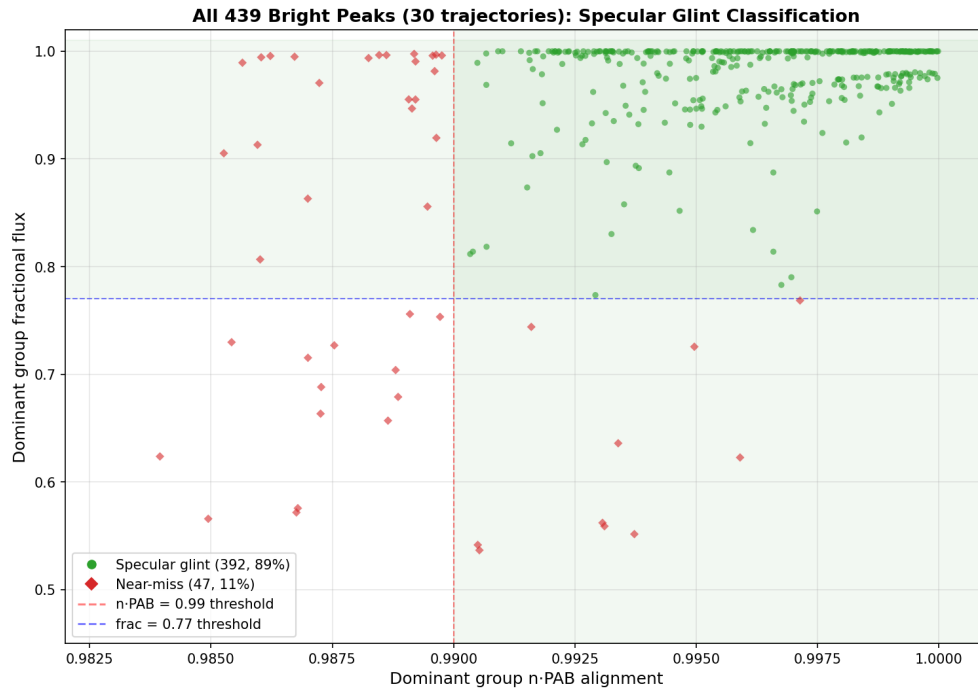


Figure 7: micro35: All 439 bright peaks. Near-misses (red) cluster near the strict thresholds.

Experiment 9: PAB Alignment as Candidate Filter

Variables	Angular threshold θ (swept 1° – 90°)
Constraints	5643 iso-brightness candidates from micro10 at epoch 183 (glint, mag 7.15); 14 unique body-frame normals from IS-901; $\text{PAB}_{\text{inertial}}$ from SPICE; baseline: 5643 random $\text{SO}(3)$ quaternions (seed 12345)
Residual	Per candidate q : $\max_g \hat{\mathbf{n}}_g \cdot [R(q) \widehat{\text{PAB}}_{\text{inertial}}]$; candidate survives if $> \cos \theta$
Solver	None — vectorised dot products (no optimisation)

Result:

Threshold	Kill (iso)	Kill (random)	Iso survivors	Truth alignment
3°	100.0%	99.1%	0	
5°	89.4%	97.6%	600	$4.62^\circ \checkmark$
8°	0.0%	93.1%	5643	

All 5643 iso-brightness candidates fall within 4.4 – 7.5° of some normal. At 5° : $10\times$ reduction while preserving truth. Key finding: iso-brightness matching at glint epochs *implicitly* selects for PAB alignment—the two constraints are partially redundant.

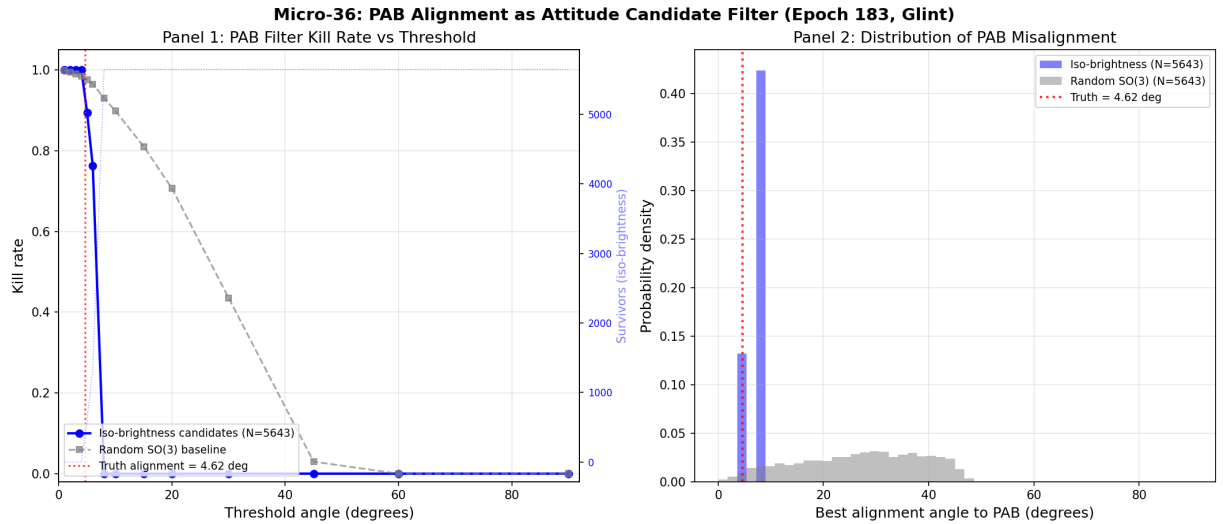


Figure 8: micro36: PAB alignment distributions. Iso-brightness candidates (blue) are already tightly clustered; random $\text{SO}(3)$ (grey) spread across the full sphere.

Experiment 10: BRDF Specular Glint Profile Characterisation

Variables	Misalignment angle θ ($0\text{--}30^\circ$, 0.1° steps); swept parameters: $n_{\text{phong}} \in \{50, 100, 200, 300, 500, 1000\}$; $r_s \in \{0.05, 0.10, 0.20, 0.40, 0.60, 0.80\}$; phase angle $\in \{5, 10, 20, 30, 45, 60\}^\circ$
Constraints	Synthetic flat plate, 1 m^2 , GEO distance (36 000 km); Ashikhmin–Shirley BRDF; $\hat{\mathbf{k}}_1, \hat{\mathbf{k}}_2$ symmetric about PAB at half-angle $= \phi/2$; facet normal swept through PAB
Residual	Apparent magnitude vs misalignment angle; FWHM of flux peak; specular = diffuse crossing angle
Solver	None — pure BRDF evaluation (no satellite, no SPICE)

Result:

n_{phong}	FWHM ($^\circ$)	Spec = Diff ($^\circ$)	Peak mag (1 m^2)
50	19.4	18.2	10.06
200	9.5	11.3	8.56
500	6.0	7.1	7.56
1000	4.3	6.0	6.81

n_{phong} controls lobe width ($\text{FWHM} \propto 1/\sqrt{n_{\text{phong}}}$). r_s controls amplitude ($\sim 4\text{ mag}$ range) but barely affects width. Phase angle is irrelevant ($< 0.01\text{ mag}$ variation across $5\text{--}60^\circ$). Area scales linearly (verified exactly).

For IS-901 ($n_{\text{phong}} = 200\text{--}267$): specular-dominated region extends to $\sim 10^\circ$.

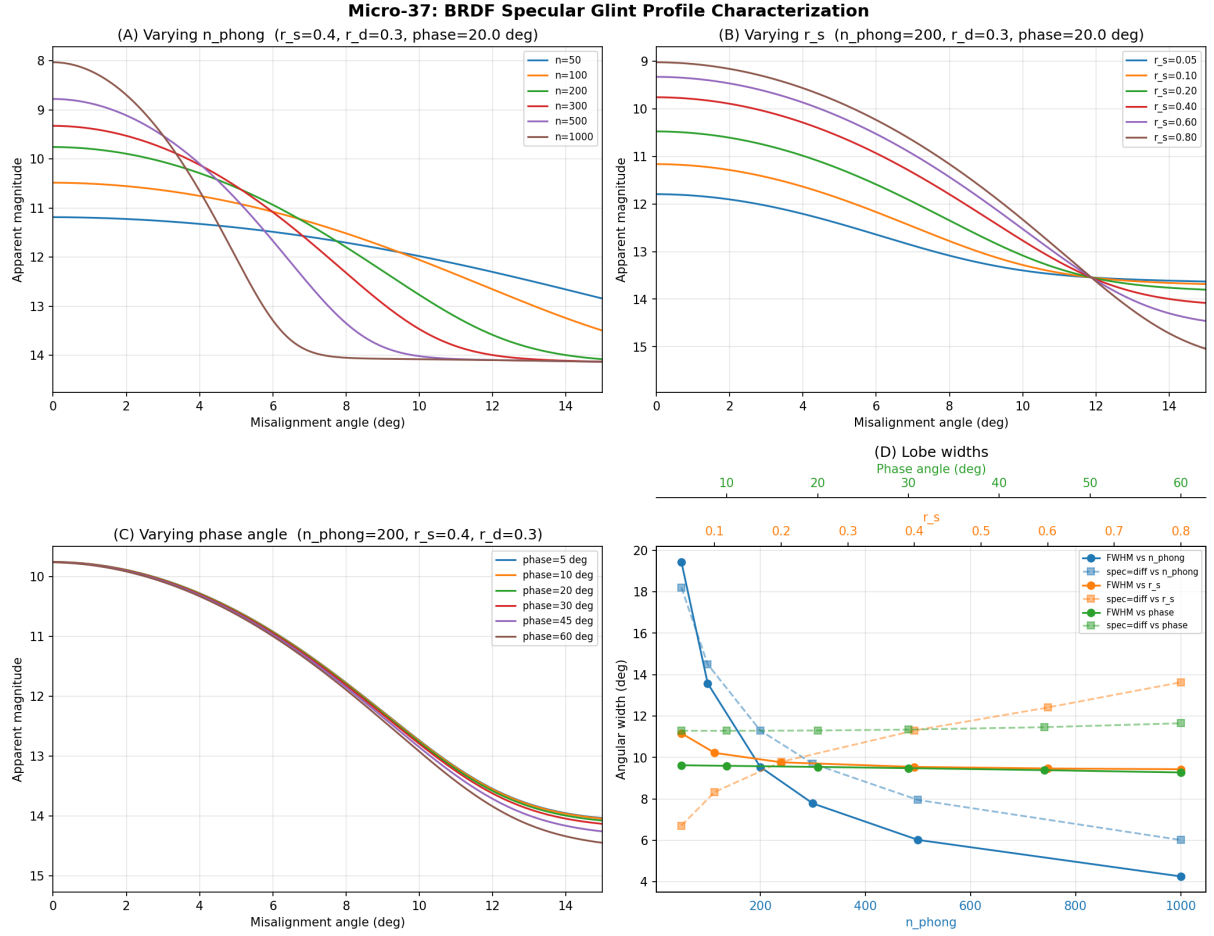


Figure 9: micro37: BRDF profile. (A) n_{phong} controls width. (B) r_s controls amplitude. (C) Phase angle irrelevant. (D) Lobe width summary.

Decisions

- Bright peaks are specular glints (single-facet, $\hat{\mathbf{n}} \cdot \text{PAB} > 0.99$). Universal across 30 trajectories.
- PAB filter gives $10\times$ candidate reduction at 5° threshold, but is partially redundant with iso-brightness matching at glint epochs.
- Specular lobe width ($\sim 9.5^\circ$ for IS-901) sets the angular scale for all alignment-based constraints.
- Phase angle has negligible effect on glint profiles—intrinsic material property.

Next Steps

1. Component identification from glint properties (brightness, duration, recurrence).
2. Multi-glint intersection: two glints from different normals $\rightarrow$ discrete (q, ω) candidates.
3. Glint-based candidate generation: enumerate PAB circle analytically per normal, replace iso-brightness sampling.
4. End-to-end pipeline test with non-oracle attitude candidates from iso-brightness stage.